# A JACOBSON UNIVERSALLY CATENARY NOETHERIAN DOMAIN WITH MAXIMAL IDEALS OF LOCAL DIMENSION 1 AND 2

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*Editorial note.* All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file examples.tex; no statement or proof step was rewritten, reordered or removed.

*Editorial note.* The counted proof is the author's complete argument: The counted proof delegates everything to the context construction and adds the two universal-catenary citations (Algebra, Morphisms - both intra-corpus); the author placeholder '(insert future reference here)' in the Claim is inside the context span and is printed verbatim.

*Editorial note.* The label lemma-Noetherian-Jacobson names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

*Editorial note.* The counted statement is carried by the author's own lemma environment in the chapter source.

*Allowed standard prerequisites (editorial).*

- Context span 1848-1917: the construction - k algebraic closure of a finite field, A = k[x,y], C = V(x) with enumerated closed points, auxiliary curves C_i, the Claim producing irreducible Z_n with {p_n} = Z_n cap (C cup C_1 cup ... cup C_n) via Cartier divisors and torsion Picard group of an affine curve, then S generated by the f_n with Z_n = V(f_n) and B = S^{-1}A, with the analysis of which maximal ideals survive (infinitely many from A, plus xB) yielding local rings of dimensions 1 and 2
- Universally catenary rings: Algebra, lemma-localization-catenary and Morphisms, lemma-ubiquity-uc

*Source reference.* Original lemma 00NJ: sources/stacks-chow/tags.txt, source lines 861–861. Prerequisite pointer invoked at 1928; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

*Source reference.* Original lemma 00KX: sources/stacks-chow/tags.txt, source lines 767–767. Prerequisite pointer invoked at 1995; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

*Source reference.* Original lemma 02JB: sources/stacks-chow/tags.txt, source lines 3330–3330. Prerequisite pointer invoked at 1929; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

*Source reference.* Original lemma 09NA: sources/stacks-chow/tags.txt, source lines 12208–12208. Prerequisite pointer invoked at 1959; it names a standard fact of a

*Date*: stacks-project:examples.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Let $k$ be the algebraic closure of a finite field. Let $A = k[x, y]$ and $X = \mathrm{Spec}(A)$. Let $C = V(x)$ be the $y$-axis (this could be any other 1-dimensional integral closed subscheme of $X$). Let $C_1, C_2, C_3, \ldots$ be an enumeration of the other integral closed subschemes of $X$ of dimension 1. Let $p_1, p_2, p_3, \ldots$ be an enumeration of the closed points of $C$.

Claim: for every $n$ there exists an irreducible closed $Z_n \subset X$ of dimension 1 such that

$$\{p_n\} = Z_n \cap (C \cup C_1 \cup \ldots \cup C_n)$$

set theoretically. To do this set $Y = C \cup C_1 \cup C_2 \cup \ldots \cup C_n$. This is a reduced affine algebraic scheme of dimension 1 over $k$. It is enough to find $f \in k[x, y]$ with $V(f) \cap Y = \{p_n\}$ set theoretically because then we can take $Z_n$ to be a suitable irreducible component of $V(f)$. Since the restriction map

$$k[x, y] \longrightarrow \Gamma(Y, \mathcal{O}_Y)$$

is surjective, it suffices to find a regular function $g$ on $Y$ whose zero set is $\{p_n\}$ set theoretically. To see this is possible, we choose an effective Cartier divisor $D \subset Y$ whose support is $p_n$ (this is possible by Varieties, Lemma 09NA). Thus it suffices to show that $\mathcal{O}_X(ND) \cong \mathcal{O}_X$ for some $N > 0$. But the Picard group of an affine 1-dimensional algebraic scheme over the algebraic closure of a finite field is torsion (insert future reference here) and we conclude the claim is true.

Choose $Z_n$ as above for all $n$. Since $k[x, y]$ is a UFD we may write $Z_n = V(f_n)$ for some irreducible element $f_n \in A$. Let $S \subset k[x, y]$ be the multiplicative subset generated by $f_1, f_2, f_3, \ldots$. Consider the Noetherian ring $B = S^{-1}A$.

Obviously, the ring map $A \to B$ identifies local rings and induces an injection $\mathrm{Spec}(B) \to \mathrm{Spec}(A)$. Moreover, looking at the curve $C_1$ we see that only the points of $C \cap C_1$ are removed when passing from $\mathrm{Spec}(A)$ to $\mathrm{Spec}(B)$. In particular, we see that $\mathrm{Spec}(B)$ has an infinite number of maximal ideals corresponding to maximal ideals of $A$. On the other hand, $xB$ is a maximal ideal because the spectrum of $B/xB$ consists of a unique prime ideal as we removed all the closed points of $C = V(x)$ (but not the generic point). Finally, for $i \geq 1$ consider the curve $C_i$. Write $C_i = V(g_i)$ for $g_i \in A$ irreducible. If $C_i = Z_n$ for some $n$, then $g_i B$ is the unit ideal. If not, then all but finitely many of the closed points of $C_i$ survive the passage from $A$ to $B$: namely, only the points of $(Z_1 \cup \ldots \cup Z_{i-1} \cup C) \cap C_i$ are removed from $C_i$.

The structure of the prime spectrum of $B$ given above shows that $B$ is Jacobson by Algebra, Lemma 00KX. The maximal ideals are the maximal ideals of $A$ which are in $\mathrm{Spec}(B)$ (and there an inifinitude of these) together with the maximal ideal $xB$. Thus we see that we have local rings of dimensions 1 and 2.

**Lemma 0.1.** *There exists a Jacobson, universally catenary, Noetherian domain $B$ with maximal ideals $\mathfrak{m}_1, \mathfrak{m}_2$ such that $\dim(B_{\mathfrak{m}_1}) = 1$ and $\dim(B_{\mathfrak{m}_2}) = 2$.*

*Proof.* The construction of $B$ is given above. We just point out that $B$ is universally catenary by Algebra, Lemma 00NJ and Morphisms, Lemma 02JB. □